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Cylicins are a structural component of the sperm calyx being indispensable for male fertility in mice and human

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Abstract

Cylicins are testis specific proteins which are exclusively expressed during spermiogenesis. In mice and humans, two Cylicins, the gonosomal X-linked Cylicin 1 (*Cylc1/CYLC1*) and the autosomal Cylicin 2 (*Cylc2/CYLC2*) genes have been identified. Cylicins are cytoskeletal proteins with an overall positive charge due to lysine-rich repeats. While Cylicins have been localized in the acrosomal region of round spermatids, they resemble a major component of the calyx within the perinuclear theca at the posterior part of mature sperm nuclei. However, the role of Cylicins during spermiogenesis has not yet been investigated. Here, we applied CRISPR/Cas9 mediated gene-editing in zygotes to establish *Cylc1*- and *Cylc2*-deficient mouse lines as a model to study the function of these proteins. *Cylc1*-deficiency resulted in male subfertility whereas *Cylc2*^{-/-}, *Cylc1*^{-/-}*Cylc2*^{+/-}, and *Cylc1*^{-/-}*Cylc2*^{-/-} males were infertile. Phenotypical characterization revealed that loss of Cylicins prevents proper calyx assembly during spermiogenesis. This results in decreased epididymal sperm counts, impaired shedding of excess cytoplasm, and severe structural malformations, ultimately resulting in impaired sperm motility. Furthermore, exome sequencing identified an infertile man with a hemizygous variant in *CYLC1* and a heterozygous variant in *CYLC2*, displaying morphological abnormalities of the sperm including the absence of the acrosome. Thus, our study highlights the relevance and importance of Cylicins for spermiogenic remodeling and male fertility in human and mouse, and provides the basis for further studies on unraveling the complex molecular interactions between perinuclear theca proteins required during spermiogenesis.

eLife assessment

This study provides **valuable** insights into the role of two under-researched sperm-specific proteins (Cylicin 1 and Cylicin 2). The authors provide **convincing** evidence that they have an essential role in sperm head structure during spermatogenesis, and that their loss leads to subfertility or infertility, with a dose-dependent phenotype. The authors identify infertile males with mutations in both Cylicin1 and Cylicin2: thus the findings from the mouse models might be applicable to understanding human male infertility with similar structural defects.

Introduction

The differentiation of round spermatids into sperm during spermiogenesis is a highly organized and spatiotemporally controlled process taking place in the seminiferous epithelium of the testis. Cellular and morphological remodeling involves DNA hypercondensation, establishment of a species-specific head morphology, removal of excess cytoplasm as well as formation of accessory structures like acrosome and flagellum. These structural changes depend on a highly efficient protein trafficking machinery and a unique sperm cytoskeleton ⁽¹⁾. One essential cytoskeletal element is the perinuclear theca (PT), which surrounds the sperm nucleus, except for the caudal edge, at the implantation site of the flagellum. The PT is supposed to serve as a structural scaffold for the sperm nucleus and resembles a rigid cytosolic protein layer, which is resistant to non-ionic detergents and high salt buffer extractions ^{(2), (3)}. The PT has been subdivided into a subacrosomal and postacrosomal part based on its localization, function, composition, and developmental origin. The subacrosomal part of the PT develops early during spermiogenesis, simultaneously with the formation of the acrosome. It is supposed to emerge from acrosomal vesicles, and presents as a thin cytosolic protein layer between the inner acrosomal membrane and the nuclear envelope ^{(4), (5)}. The postacrosomal part of the PT, also known as the sperm calyx, originates from cytosolic proteins that are transported via the manchette and assembled during sperm head elongation ⁽⁴⁾. It is located in between the nuclear envelope and the sperm plasma membrane.

Proteomic analyses using murine and bovine PT extracts revealed 500-800 different proteins, highlighting its high molecular complexity ^{(6), (7)}. Apart from cytoskeletal proteins, signaling molecules and several *de novo* synthesized core histones were identified. They are proposed to be essential for sperm-egg interaction and chromatin remodeling of the male pronucleus at early post-fertilization stages ^{(8), (9), (4), (10), (11)}. Many structural proteins of the PT are testis-specific and uniquely expressed in the PT, including Calicin (*Ccin*) ^{(12), (2), (13)}, Cylicin 1 (*Cylc1*) and Cylicin 2 (*Cylc2*) ^{(2), (14), (15)}, actin-capping proteins CPβ3 and CPα3 ^{(16)–(18)} as well as actin-related proteins Arp-T1 and Arp-T2 ⁽¹⁹⁾. Despite being characterized on molecular level, their function remains largely elusive.

Both, *Ccin* and Cylicins are highly basic proteins with a predominant localization in the calyx of mature sperm ^{(14), (15), (12)}. In most species, two Cylicin genes, *Cylc1* and *Cylc2*, have been identified. They are characterized by repetitive lysine-lysine-aspartic acid (KKD) and lysin-lysine-glutamic acid (KKE) peptide motifs, resulting in an isoelectric point (IEP) > pH 10 ^{(14), (15)}. While the *CYLC2/Cylc2* gene is encoded on autosomes, the human and murine *CYLC1/Cylc1* gene is encoded on the X-chromosome, resulting in hemizyosity in males.

Further, in bovine, *Cylc2* serves as actin-binding protein of the sperm perinuclear cytoskeleton ⁽²⁰⁾. Cylicins seem to be cytoskeletal regulators and required for proper sperm head architecture.

In this study, we report the CRISPR/Cas9-mediated generation and characterization of *Cylc1*-, *Cylc2*-, and *Cylc1/2*-deficient mice, demonstrating that Cylicins are indispensable for male fertility and play a key function in the formation of the sperm calyx. We show that Cylicins are required for the maintenance of the PT integrity during spermiogenesis and in mature sperm. Their deficiency results in morphological defects of the sperm head, acrosome, and mid-piece. Our analyses revealed that *Cylicin* genes are evolutionary conserved in rodents and primates. Furthermore, we identified Cylicin variants in an infertile man, demonstrating the conserved role of Cylicins in regulating male infertility.

Results

Cylicins are indispensable for male fertility in mice

Cylicins were first discovered and characterized in the early nineties, and a role in sperm head architecture was postulated due to their subcellular localization ^{(2), (14), (15)}, however, any functional evidence has been lacking. To address the question of the role of Cylicins during spermiogenesis, we used CRISPR/Cas9-mediated gene-editing to generate *Cylc1* and *Cylc2* deficient mouse models. First, for *Cylc1*, a pair of sgRNAs targeting exon 5 was designed, and a mouse line with a frameshift inducing deletion of 1.585 kb, accounting for 85% of the *Cylc1* coding sequence, was established. Next, two guide RNAs targeting exon 4/5 of the *Cylc2* gene were applied to establish a *Cylc2* deficient mouse line with a 1.145 kb frameshift inducing deletion (Fig. 1 A). In both lines, the majority of predicted functional domains with repetitive, lysine-lysine-aspartic acid (KKD) and lysin-lysin-glutamic acid (KKE) peptide motifs were depleted. Depletion was confirmed by PCR-based genotyping (Fig. 1 B).

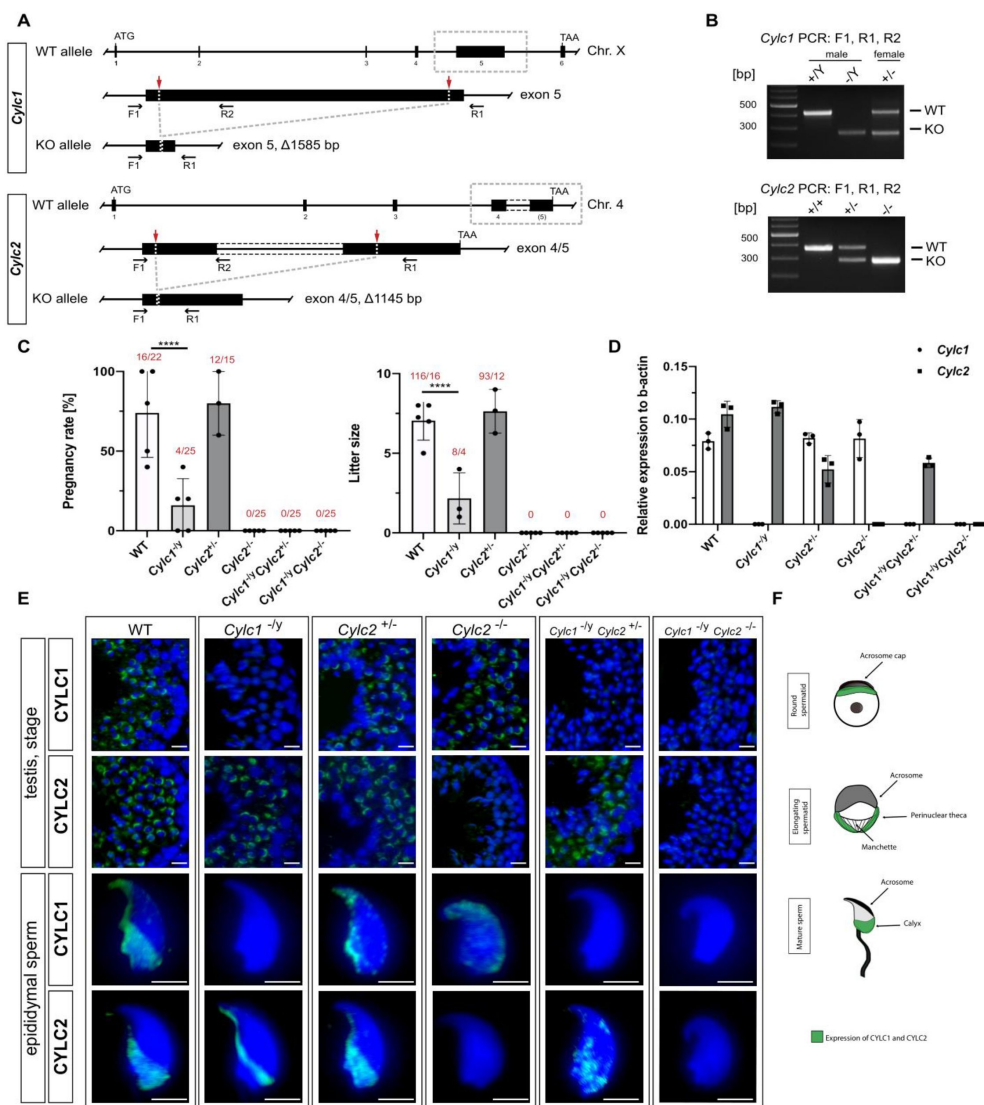


Figure 1:

Loss of *Cylc1* or *Cylc2* results in impaired male fertility.

(A) Schematic representation of the *Cylc1* and *Cylc2* gene structure and targeting strategy for CRISPR/Cas9 mediated generation of *Cylc1*- and *Cylc2*- deficient alleles. Targeting sites of guide RNAs are depicted by red arrows. Genotyping primer binding sites are depicted by black arrows.

(B) Representative genotyping PCR of *Cylc1*- and *Cylc2*- deficient mice.

(C) Fertility analysis of Cyclin-deficient mice visualized by mean litter size and pregnancy rate (%) in comparison to wild type matings. Black dots represent mean values obtained for each male included in fertility testing. Columns represent mean values $\pm$ standard deviation (SD). Total number

of offspring per total number of pregnancies as well as total number of pregnancies per total number of plugs are depicted above each bar.

(D) Expression of *Cylc1* and *Cylc2* in testicular tissue of WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-}*Cylc2*^{+/-} and *Cylc1*^{-/-}*Cylc2*^{-/-} mice analyzed by qRT-PCR.

(E) Immunofluorescent staining of testicular tissue and cauda epididymal sperm from WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-}*Cylc2*^{+/-} and *Cylc1*^{-/-}*Cylc2*^{-/-} males against CYLC1 and CYLC2. Cell nuclei were counterstained with DAPI. Scale bar: 5 μ m.

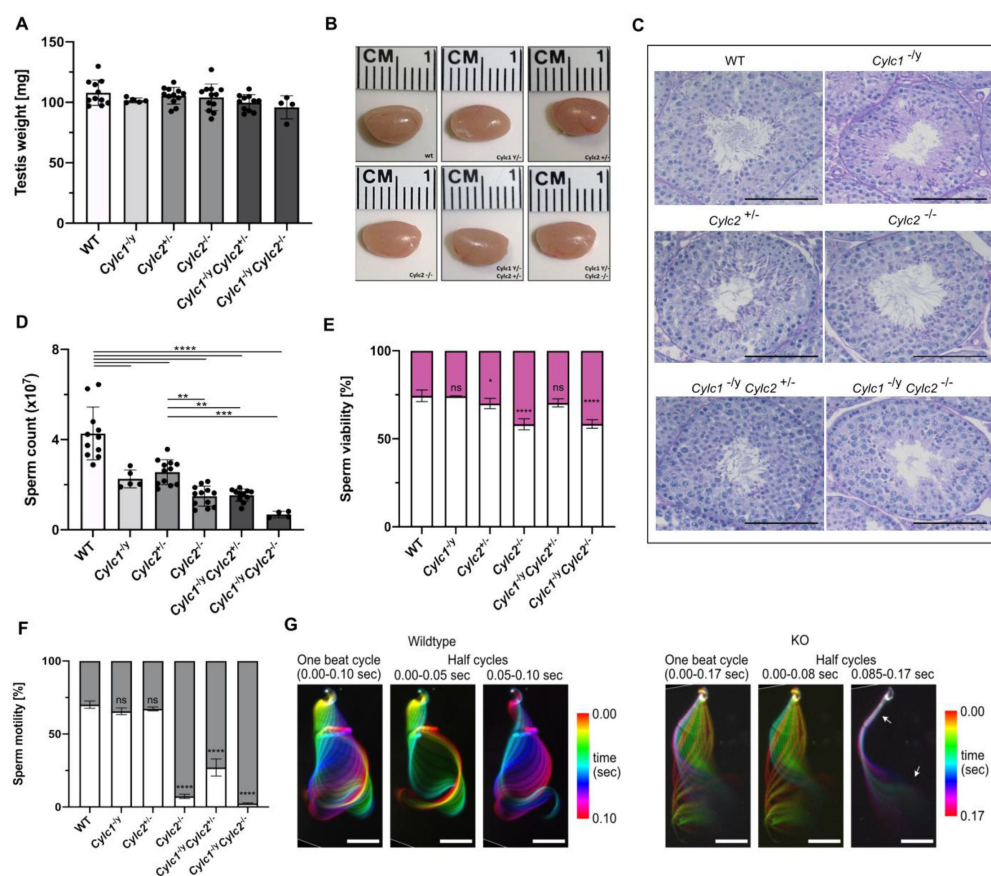
Fertility testing of *Cylc1*^{-/-} males revealed significantly reduced pregnancy rates (16 %) and mean litter size (2.2) (Fig. 1 C). *Cylc2*^{-/-} males were infertile, while *Cylc2*^{+/-} males showed no significant difference in fertility parameters compared to wildtype mice (Fig. 1 C). Additionally, established mouse lines were intercrossed to generate *Cylc1*^{-/-}*Cylc2*^{+/-} and *Cylc1*^{-/-}*Cylc2*^{-/-} males. Of note, *Cylc1*^{-/-}*Cylc2*^{+/-}, and *Cylc1*^{-/-}*Cylc2*^{-/-} males were infertile (Fig. 1 C). This indicates that loss of *Cylc1* alone is partially tolerated, as suggested by subfertility of *Cylc1*^{-/-} males, whereas the additional loss of one *Cylc2* allele renders male mice infertile.

Taken together, the results suggest, that two functional Cylicin alleles are required for male fertility.

qRT PCR confirmed the absence of *Cylc1* and/or *Cylc2* transcripts in Cylicin-deficient animals (Fig. 1 D). For *Cylc2*^{+/-} animals a gene-dosage effect was observed, as indicated by a 50 % reduced *Cylc2* expression. Neither loss of *Cylc1* nor *Cylc2* resulted in upregulation of *Cylc2* or *Cylc1*, respectively.

Next, due to the lack of commercial antisera, polyclonal antibodies against murine CYLC1 and CYLC2 were raised to visualize the localization of Cylicins during spermiogenesis. Immunofluorescence staining of wildtype testicular tissue showed presence of both, CYLC1 and CYLC2 from the round spermatid stage onward (Fig. 1 D, E). The signal was first detectable in the subacrosomal region as a cap like structure, lining the developing acrosome (Fig. 1 D, E). As the spermatids elongate, CYLC1 and CYLC2 move across the PT towards the caudal part of the cell (Fig. S1 A). At later steps of spermiogenesis, the localization in the subacrosomal part of the PT faded while it intensified in the postacrosomal calyx region. (Fig. 1 D, E). Of note the localization of CYLC1 and CYLC2 in the calyx of mature sperm has been reported in bovine and human ^{(14), (15), (20)}. The generated antibodies did not stain testicular tissue and mature sperm of *Cylc1*- and *Cylc2*-deficient males demonstrating i) specificity of the antibodies and ii) validating the knockout mice (Fig. 1 E).

Next, spermiogenesis of Cylicin-deficient males was analyzed in detail. Gross testicular morphology as well as testicular weight were not significantly altered (Fig. 2 A, B). The testicular morphology appeared unaltered, with all stages of spermatogenesis being detectable in PAS-stained testicular sections (Fig. 2 C). However, a strong decline in cauda epididymal sperm counts was observed in all Cylicin-deficient males. For *Cylc1*^{-/-} and *Cylc2*^{+/-} males, a moderate reduction of 40-47% was determined, whereas *Cylc2*^{-/-} and *Cylc1*^{-/-}*Cylc2*^{+/-} displayed an approx. 65% reduction in epididymal sperm counts compared to wildtype mice (Fig. 2 D). In *Cylc1*^{-/-}*Cylc2*^{-/-} males, spermiogenesis was most impaired, as indicated by an 85% reduction of the sperm count (Fig. 2 D). Eosin-nigrosine staining revealed that the viability of epididymal sperm from all genotypes was above the lower reference limit of 58% (Fig. 2 E, Fig. S1 B). However, viability of *Cylc2*^{-/-} and *Cylc1*^{-/-}*Cylc2*^{-/-} sperm was reduced by approx. 15% compared to wildtype sperm. (Fig. 2 E). The motility of *Cylc1*^{-/-} and *Cylc2*^{+/-} sperm remained unchanged compared to wildtype sperm (around 60% motile cells), whereas motility of *Cylc2*^{-/-} sperm was drastically reduced to only 7% motile sperm (Fig. 2 F) and the motility of *Cylc1*^{-/-}*Cylc2*^{-/-} sperm was reduced to 2% motile sperm. In addition, the few motile sperm cells were not progressive but were swimming in circular trajectories. Interestingly, in *Cylc1*^{-/-}*Cylc2*^{+/-} mice, sperm motility was reduced as well, but less drastically, with 27% of sperm cells being motile (Fig. 2 F).



bar: 100 μ m.

(D) Epididymal sperm count ($\times 10^7$) of WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-}*Cylc2*^{+/-} and *Cylc1*^{-/-}*Cylc2*^{-/-} males. Mean values $\pm$ SD are shown; block dots represent data points for individual males.

(E) Viability of the epididymal sperm stained with Eosin Nigrosine. Percentage of Eosin negative (viable) and Eosin positive (inviable) sperm is shown. Data represented as mean $\pm$ SD.

(F) Motility of the epididymal sperm of WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-}*Cylc2*^{+/-} and *Cylc1*^{-/-}*Cylc2*^{-/-} males activated in TYH medium.

(G) Full and half-beat cycle plots of the flagellar beat are shown for WT and *Cylc2*^{-/-} spermatozoa. Half beat cycle shows the stiffness of the midpiece (upper arrow) and high oscillations (lower arrow) in *Cylc2*^{-/-} sperm in one direction of the beat.

Cylc2^{-/-} sperm cells have altered flagellar beat

The SpermQ software was used to analyze the flagellar beat of *Cylc2*^{-/-} sperm in detail (21). *Cylc2*^{-/-} sperm showed stiffness in the neck and a reduced amplitude of the initial flagellar beat, as well as reduced average curvature of the flagellum during a single beat (Fig. S4 A). Interestingly, we observed that the flagellar beat of *Cylc2*^{-/-} sperm cells was similar to wildtype cells with - however - interruptions during which midpiece and initial principal piece appeared stiff whereas high frequency beating occurs at the flagellar tip (Fig. 2G,

Figure 2:

Loss of Cylicins impairs spermiogenesis

(A) Testis weight [mg] and sperm count ($\times 10^7$) of WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-}*Cylc2*^{+/-} and *Cylc1*^{-/-}*Cylc2*^{-/-} males. Mean values $\pm$ SD are shown; block dots represent data points for individual males.

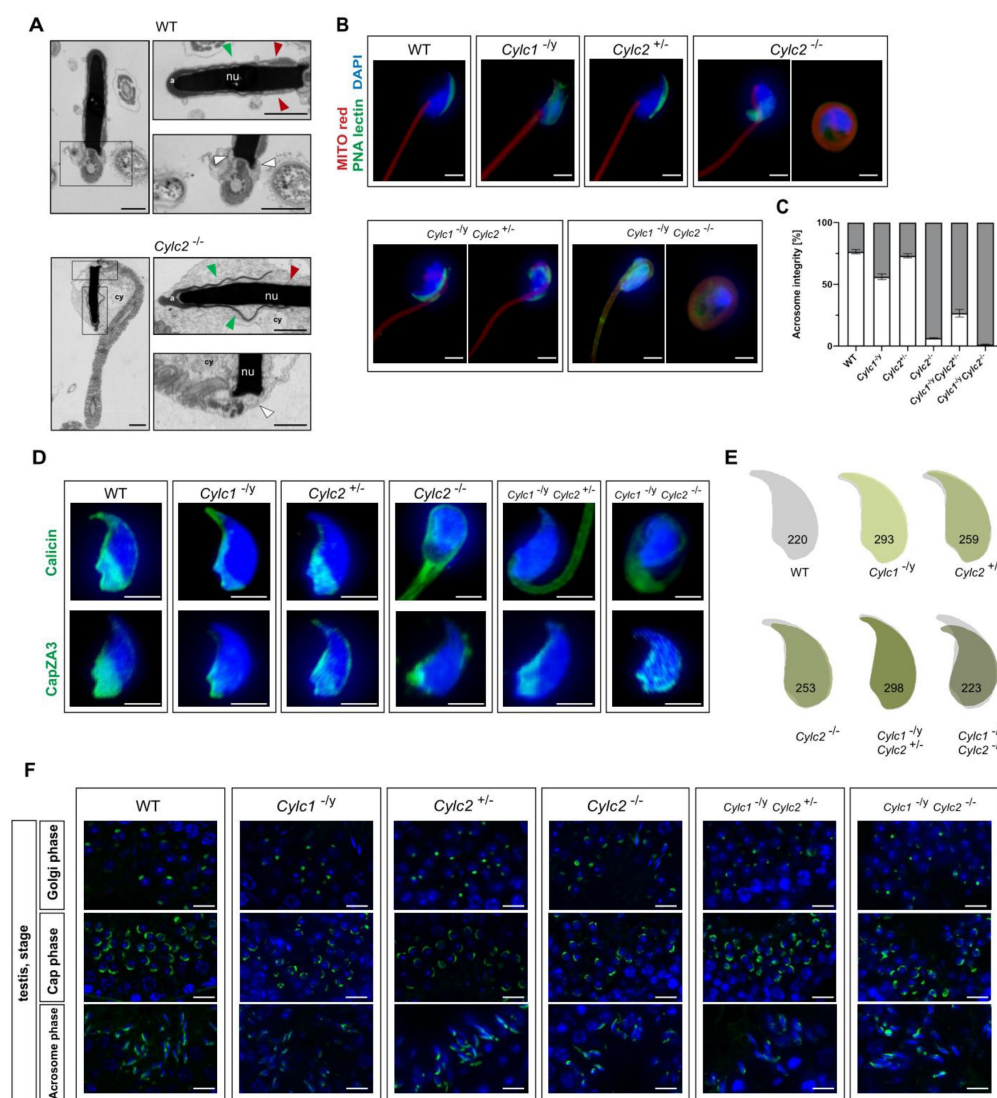
(B) Comparable photographs of the testes of WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-}*Cylc2*^{+/-} and *Cylc1*^{-/-}*Cylc2*^{-/-} mice.

(C) PAS staining of testicular sections from WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-}*Cylc2*^{+/-} and *Cylc1*^{-/-}*Cylc2*^{-/-} mice. Scale

supplementary Video 1, supplementary Video 2). These interruptions occurred only on the open-hook side and the duration of such interruptions varied from beat to beat. As these anomalies introduce a high asymmetry to the flagellar beating, swimming of such sperm is very likely perturbed limiting the capability to reach the egg⁽²²⁾. Of note similar phenotypes have been observed for sperm with structural defects in the axoneme⁽²²⁾. Thus, we hypothesized that there may be structural defects in *Cylc2*-deficient sperm.

Sperm morphology is severely altered in Cylicin deficient mice

Indeed, transmission electron microscopy (TEM) of epididymal sperm revealed that *Cylc2*-deficiency causes severe structural defects (Fig. 3 A, Fig. S2 A): *Cylc2*^{-/-} sperm showed coiling of the tail and dislocation of the tail connecting piece from the basal plate, resulting in parallel position of head and tail (Fig. 3 A, white arrowheads). The coiling of the flagellum was also observed using bright field microscopy (Fig. S2 B). Furthermore, in *Cylc2*^{-/-} sperm, excess of cytoplasm was observed, located around the nucleus and coiled tail (Fig. 3 A). Anomalies of the head were observed at the level of the PT, while the nuclei appeared unaltered. In all *Cylc2*^{-/-} sperm cells, the posterior portion of PT – calyx was absent (Fig. 3 A, red arrowheads). Instead of surrounding the nucleus entirely, the PT in *Cylc2*^{-/-} sperm appeared interrupted, missing completely its caudal part. Further, we observed loosening of the peri-acrosomal region, which is not compact and adherent to the nucleus (Fig. 3 A, green arrowheads). Although the axoneme structure appeared unaltered, presenting typical 9+2 microtubular composition, many *Cylc2*^{-/-} sperm cells had damaged mitochondria (Fig. S2 A).



Cycl1^{-/-}*Cycl2*^{+/-} and *Cycl1*^{-/-}*Cycl2*^{-/-} mice is shown. Acrosome aberrations and tail coiling were counted.

(D) Representative pictures of immunofluorescent staining against PT proteins CCIN (upper panel) and CAPZA3 (lower panel) in WT, *Cycl1*^{-/-}, *Cycl2*^{+/-}, *Cycl2*^{-/-}, *Cycl1*^{-/-}*Cycl2*^{+/-} and *Cycl1*^{-/-}*Cycl2*^{-/-} sperm. Nuclei were counterstained with DAPI. Scale bar: 5 μ m.

(E) Nuclear morphology analysis of WT, *Cycl1*^{-/-}, *Cycl2*^{+/-}, *Cycl2*^{-/-}, *Cycl1*^{-/-}*Cycl2*^{+/-} and *Cycl1*^{-/-}*Cycl2*^{-/-} sperm. Number of cells analyzed for each genotype is shown.

PNA-FITC lectin immunofluorescence staining of the acrosome in testicular tissue of WT, *Cycl1*^{-/-}, *Cycl2*^{+/-}, *Cycl2*^{-/-}, *Cycl1*^{-/-}*Cycl2*^{+/-} and *Cycl1*^{-/-}*Cycl2*^{-/-} mice (green). Golgi phase of acrosome biogenesis at round spermatid stage (I-IV) is visible in the upper panel. Middle panel shows cap phases on round spermatids (stage V-VIII). In the bottom panel acrosomal phase is shown (stage IX-XI). Nuclei were counterstained with DAPI. Scale bar: 10 μ m.

Next, we used PNA-FITC lectin immunofluorescence staining to analyze acrosome localization in mature sperm, as well as MITOred to visualize mitochondria in the flagellum (Fig. 3 B, C). Loss of *Cycl1* alone caused malformations of the acrosome in around 42% of

mature sperm, while their flagellum appeared unaltered and properly connected to the head. While *Cylc2*^{+/-} males showed normal sperm morphology, 94% of *Cylc2*^{-/-} mature sperm cells displayed morphological alterations of head and mid-piece (Fig. 3 B, C). *Cylc2*^{-/-} sperm cells showed acrosome malformations, bending of the neck region, and/or coiling of the flagellum, occasionally resulting in its wrapping around the sperm head (Fig. 3 B). While 75% of *Cylc1*^{-/-}*Cylc2*^{+/-} sperm showed these morphological alterations, 99% of *Cylc1*^{-/-}*Cylc2*^{-/-} sperm presented with coiled tail and abnormal acrosome (Fig. 3 B, C).

Loss of Cylicins prohibits calyx assembly during spermiogenesis

Since TEM micrographs revealed absence of calyx in *Cylc2*^{-/-} epididymal sperm (Fig. 3 A, red arrowheads), we analyzed the localization of other calyx specific proteins, such as CCIN and CapZa3. In *Cylc1*^{-/-} and *Cylc2*^{+/-} sperm, CCIN localization remained unchanged, being present in the calyx and in the ventral portion of PT as described previously (7). However, in *Cylc2*^{-/-} sperm, CCIN localizes to the tail or in random parts of the sperm head (Fig. 3 D). In *Cylc1*^{-/-}*Cylc2*^{+/-} and *Cylc1*^{-/-}*Cylc2*^{-/-} sperm, the localization of CCIN was also altered, with the signal mainly being present in the sperm tail. CapZa3 forms a heterodimer with CapZβ3, creating a functional complex that localizes in the calyx (23). Immunofluorescence stainings revealed that the localization of CapZa3 remained unchanged in *Cylc1*^{-/-} and *Cylc2*^{+/-} mice compared to wildtype mice. In many *Cylc2*^{-/-} sperm cells, CapZa3 localizes in the caudal portion of the head but without regular calyx localization (Fig. 3 D). Interestingly, *Cylc1*^{-/-}*Cylc2*^{+/-} mice showed less severe anomalies of the calyx and although CCIN is located almost exclusively in the tail, CapZa3 maintains the correct calyx localization (Fig. 3 D). Finally, *Cylc1*^{-/-}*Cylc2*^{-/-} spermatozoa showed CapZa3 localization across the sperm head without regular calyx shape. These results suggest that the loss of Cylicins impairs the formation of calyx and the correct localization of its components, which might cause further morphological anomalies of the sperm (Fig. 3 D).

Loss of *Cylc2* impairs sperm nuclear morphology

To analyze in detail the sperm head, we used Nuclear Morphology software on DAPI stained sperm samples. *Cylc1*^{-/-} and *Cylc2*^{+/-} sperm showed no alterations of the nuclear shape when compared to wildtype (Fig. 3 E, Fig. S2C, D). However, heads of *Cylc2*^{-/-} and *Cylc1*^{-/-}*Cylc2*^{-/-} sperm appeared smaller, with shorter hooks and increased circularity of the nuclei as well as reduced elongation (Fig. 3 E, Fig. S2D). This indicates that loss of *Cylc2* causes a mild globozoospermia. Interestingly, *Cylc1*^{-/-}*Cylc2*^{+/-} sperm heads were not drastically altered, suggesting that only *Cylc2* has a crucial role for sperm head shaping, and one functional *Cylc2* allele is sufficient to maintain the correct shape of the nucleus (Fig. 3 E, Fig. S2D). Furthermore, the head of many *Cylc2*-deficient sperm appeared less concave at the region proximal to the connecting piece (Fig. 3 E).

Cylicins are required for acrosome attachment to the nuclear envelope

PNA lectin staining of testicular tissue sections was performed to investigate acrosome biogenesis. During Golgi phase, the acrosome first starts to appear as an aggregation of proacrosomal vesicles into a single granule. This premature acrosomal structure was unaltered in all genotypes, with PNA signal appearing as a small dot on one pole of round spermatids (Fig. 3 F). During cap phase, acrosome grows to cover the apical part of the nucleus. In wildtype and *Cylc2*^{+/-} mice, the forming acrosome appears equally smooth and

shows a regular cap structure on the perinuclear region of round spermatids. However, in some of the round spermatids from *Cylc2*^{-/-} and *Cylc1*^{-/-} mice, gaps in the forming acrosome are observed, as well as an irregular shape of the cap. In *Cylc1*^{-/-}*Cylc2*^{+/-} and *Cylc1*^{-/-}*Cylc2*^{-/-} mice, most of the round spermatids were deformed or displayed irregularly localized caps (Fig. 3 F, Fig. S3A). At acrosome phase, many elongating spermatids of *Cylc1*^{-/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-}*Cylc2*^{+/-} and *Cylc2*^{-/-}*Cylc1*^{-/-} mice have irregular acrosome (Fig. 3 F, Fig. S3A). Detachment of the acrosome from the nuclear envelope is evident in testis samples of *Cylc2*^{-/-} and *Cylc1*^{-/-}*Cylc2*^{-/-} male mice. These results suggest that Cylicins are required for the attachment of the developing acrosome to the nuclear envelope during spermiogenesis.

Cylicin-deficiency results in abnormal manchette elongation and disassembly

Next, we investigated the role of Cylicins during formation and development of the manchette – a sperm-specific, transient structure that represents a microtubular platform for protein transport. Transport of the intracellular vesicles is crucial for the formation of the flagellum, acrosome assembly and removal of excess cytoplasm. The manchette is first detected at step 8 at the perinuclear ring of round spermatids, just prior to their elongation. During the next steps of spermiogenesis, as the spermatids elongate, manchette moves towards the neck region in a skirt-like structure and starts disassembling at step 13 when the elongation is complete⁽²⁴⁾. We used immunofluorescence staining of acetylated α -tubulin on squash testis samples containing spermatids at different stages of spermiogenesis to investigate whether the altered head shape, calyx structure, and tail-head connection anomalies originate from possible defects of the manchette structure. Spermatids starting from stage 8 were observed individually for step-to-step comparison. In all genotypes, a cap-like shape of the manchette was observed in step 8 round spermatids, suggesting that the manchette assembles properly and starts elongating towards the neck region during step 9 (Fig. S3 B). In all genotypes, the typical skirt-like structure was observed at the caudal region of the spermatids at step 10 and 11, suggesting that the manchette assembles correctly even in Cylicin-deficient sperm (Fig. 4 A). In spermatids from *Cylc1*^{-/-} and *Cylc2*^{+/-} mice, regular manchette development was observed in further steps of spermiogenesis (Fig. 4 A). However, starting from step 12, spermatids from *Cylc2*^{-/-}, *Cylc1*^{-/-}*Cylc2*^{+/-}, and *Cylc1*^{-/-}*Cylc2*^{-/-} mice showed abnormal manchette elongation, which became more prominent at step 13 (Fig. 4 A). Interestingly, some of *Cylc2*-deficient spermatids showed shifting of the manchette to the ventral side of the nucleus along with excessive elongation. At step 16, the manchette was normally disassembled in wildtype, *Cylc1*^{-/-}, and *Cylc2*^{+/-} spermatids (Fig. 4 A). However, *Cylc2*^{-/-}, *Cylc1*^{-/-}*Cylc2*^{+/-}, and *Cylc1*^{-/-}*Cylc2*^{-/-} spermatids showed a persistent α -tubulin signal, indicating that disassembly of the manchette is delayed or incomplete (Fig. 4 A).

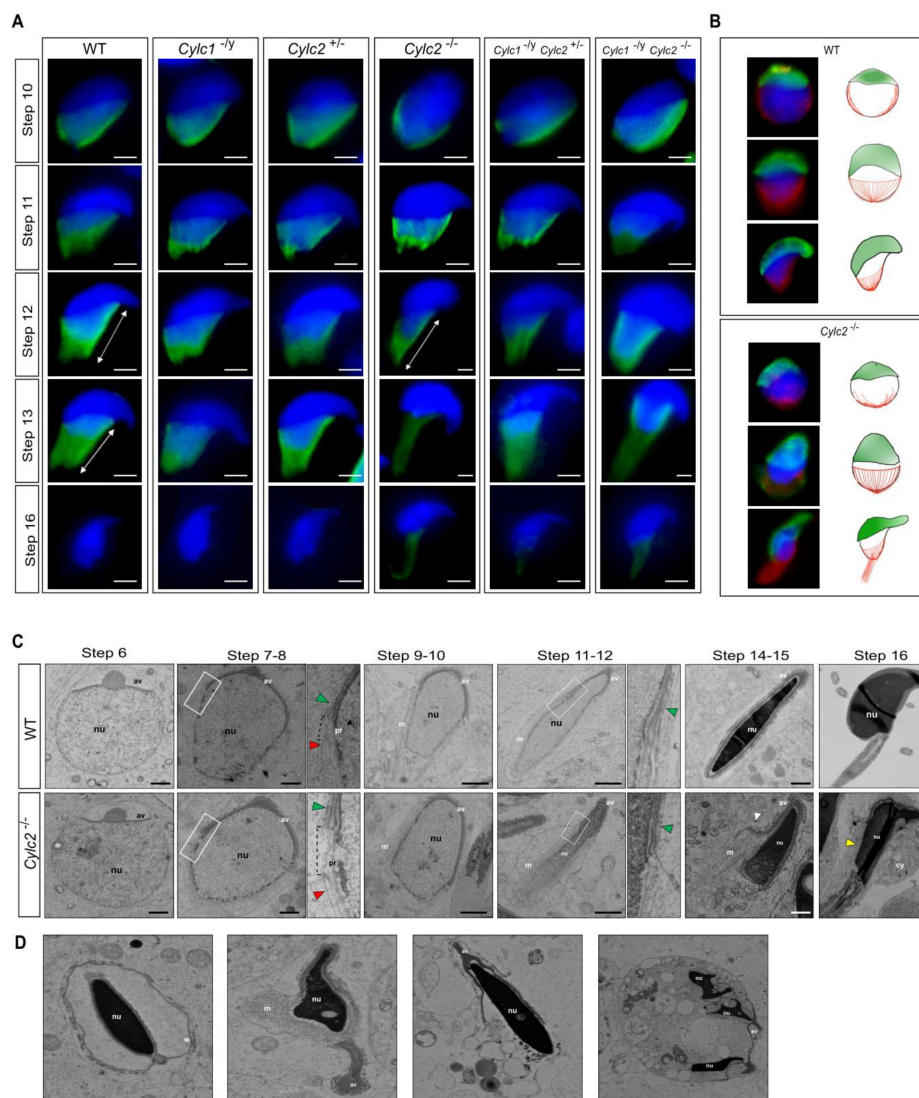


Figure 4:

***Cylc2* deficiency causes delay in manchette removal**

(A) Immunofluorescence staining of Tubulin manchette structure in the squash testis samples of WT, *Cylc1*^{-/-}, *Cylc2*^{+/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-}*Cylc2*^{+/-} and *Cylc1*^{-/-}*Cylc2*^{-/-} mice. Spermatids in different stages of spermiogenesis were shown, for step-to-step comparison Scale bar: 5 μ m.

(B) Co-staining of the manchette with HOOK1 and acrosome with PNA-lectin is shown in round, elongating and elongated spermatids of WT (upper panel) and *Cylc2*^{-/-} mice (lower panel). Schematic representation shows acrosomal structure (green) and manchette filaments (red).

(C) TEM micrographs of testicular tissues of WT and *Cylc2*^{-/-} mice. Single spermatids from step 6 to step 16 are shown. nu: nucleus; av: acrosomal vesicle; pr: perinuclear ring; m: manchette microtubules cy: cytoplasm Scale bar: 1 μ m.

μ m.

(D) TEM micrographs of degrading damaged spermatids in testicular sections of *Cylc2*^{-/-} mice. Scale bar: 5 μ m.

Other than α -tubulin, we also used HOOK1 as manchette marker. HOOK1 is a member of a family of coiled-coil proteins, which bind to microtubules and organelles and regulate microtubule trafficking during endocytosis and spermiogenesis⁽²⁵⁾. Co-staining of the spermatids with antibodies against PNA lectin and HOOK1 revealed that abnormal manchette elongation and acrosome anomalies simultaneously occurred in elongating spermatids of *Cylc2*^{-/-} male mice (Fig. 4 B): While manchette was moving excessively towards the tail, the developing acrosome lost its compact adherence to the nuclear envelope (Fig. 4 B), causing the concomitant anomalies in the mature sperm. Schematic representation shows acrosome biogenesis and manchette development in wildtype and *Cylc2*^{-/-} spermatids (Fig. 4 B). While round spermatids of *Cylc2*^{-/-} mice elongated as those of the wildtype sperm, the manchette elongated abnormally and the acrosome became loosened (Fig. 4 B).

Transmission electron microscopy of testicular tissues of wildtype and *Cylc2*^{-/-} mice showed detailed structure of spermatids at different steps of spermiogenesis (Fig. 4 C). Round spermatids at step 6 showed correct assembly of the proacrosomal granule typical for Golgi phase, as well as normal looking PT in both wildtype and *Cylc2*^{-/-} mice (Fig. 4 C). However, in *Cylc2*^{-/-} elongating spermatids already at step 8, we observed wide gaps in the PT at the perinuclear ring level, between the growing acrosome (Fig. 4 C, green arrowheads) and the developing manchette (Fig. 4 C, red arrowheads; gaps are depicted by dashed line). In wildtype spermatids, these structures were in close contact (Fig. 4 C). Acrosome biogenesis proceeded regularly with acrosomal caps forming at the apical pole of the spermatids. At step 9-10 spermatids of *Cylc2*^{-/-} mice we already observed manchette anomalies. In many cells, microtubules appeared longer on one side of the nucleus than on the other, displacing the acrosome to the side. In turn, this might result in abnormal attachment of the developing flagellum to the basal plate and consequently flipping of the head and coiling of the tail in mature spermatozoa (Fig. 4 C). At step 11-12, excessive elongation of the manchette in *Cylc2*^{-/-} spermatids as well as the gaps in PT were observed. Whereas elongated spermatids at step 14-15 in wildtype sperm already disassembled their manchette and the PT appeared as a unique structure that compactly surrounds nucleus, in *Cylc2*^{-/-} spermatids, remaining microtubules failed to disassemble, and the acrosome detached from the nuclear envelope. In addition, at step 16, the calyx was absent, and an excess of cytoplasm surrounded the nucleus and flagellum (Fig. 4 C, yellow arrowhead). Furthermore, many damaged and degrading cells were observed in testicular tissue TEM samples, having perforated nuclei and detached structures (Fig. 4 D). Interestingly, phagosomes with cellular remains were observed far away from the lumen and sometimes even at the basal membrane of the tubuli, suggesting that the cells that suffer most severe structural damage are being removed. This mechanism of removing malformed cells explains the reduction of epididymal sperm count in Cylicin-deficient genotypes.

We conclude that loss of two Cylicin alleles does not disturb the initial phase of manchette formation but causes the delocalization of the manchette towards the tail and abolishes timely manchette disassembly. The localization of manchette on the caudal portion of spermatids coincides with the localization of the calyx in the mature sperm, so these results indicate that the manchette might be maintained longer to compensate for the missing formation of the calyx structure.

Cylc2* coding sequence is slightly more conserved among species than *Cylc1

To address why *Cylc2* deficiency causes more severe phenotypic alterations than *Cylc1*-deficiency in mice, we performed evolutionary analysis of both genes. Selective pressures on protein level were represented by calculation of the nonsynonymous/synonymous substitution rate ratio ($\omega = dN/dS$). It distinguishes between purifying selection ($\omega < 1$), neutral evolution ($\omega = 1$) and positive selection ($\omega > 1$). Analysis of the selective constraints on *Cylc1* and *Cylc2* across rodents and primates revealed that both genes' coding sequences are under purifying selection in general, although conservation is weaker in *Cylc1* trending towards a more relaxed constraint (Fig. S4 A, Table S1). For *Cylc1*, the evolutionary rate (ω) of the whole sequence for the whole tree analyzed was significantly lower than, but close to 1 at $\omega = 0.85$. For *Cylc2*, it was lower at $\omega = 0.49$ (see Table S1). A model allowing for separate calculation of the evolutionary rate for primates and rodents, did not detect a significant difference between the clades, neither for *Cylc1* nor for *Cylc2*, indicating that the sequences are equally conserved in both clades.

To analyze the selective pressure across the coding sequence in more detail, we calculated the evolutionary rates for each codon site across the whole tree. The model allowing for positive selection on codon sites explained the data significantly better than the null model according to likelihood ratio analysis (see Table S1). For *Cylc1*, 34% of codon sites were assigned to the conserved site class, 51% to the neutral/relaxed constraint site class, and 15% to the positive selection site class. For *Cylc2*, 47% of codon sites were assigned to the conserved site class, 44% to the neutral/relaxed constraint site class, and 9% to the positive selection site class. Of the codon sites assigned to the positive selection site class three sites were significantly positively selected for *Cylc1* and 11 for *Cylc2* (Table S1). The distribution of evolutionary rates across the sequence revealed largely relaxed constraint in the N-terminal region with several sites of positive selection and purifying selection with positive selection hot spots in the C-terminal region of *Cylc1*. The N-terminal region of *Cylc2* seems to be largely conserved with a pattern of conservation and stronger positive selection in the internal and C-terminal region. Interestingly, codon sites encoding lysine residues, which are proposed to be of functional importance for Cylicins, are mostly conserved. For *Cylc1*, 17% of lysine residues are significantly conserved and 35% of significantly conserved codons encode for lysine. For *Cylc2*, this pattern is even more pronounced with 27.9% of lysine codons being significantly conserved and 24.3% of all conserved sites encoding for lysine (Fig. S4).

Cylicins are required for normal sperm morphology in human

As loss of two Cylicin alleles causes fertility defects in mice, we next addressed whether infertile men also display variants in *CYLC1/CYLC2*.

Exome sequencing within the MERGE (**M**ale **R**eproductive **G**enomics study) cohort identified one patient (M2270) carrying rare (MAF < 0.01, gnomAD) missense variants in both *CYLC1* and *CYLC2*. The hemizygous variant c.1720G>C in *CYLC1* leads to an amino acid exchange from glutamic acid to glutamine (p.(Glu574Gln)) which is predicted to be deleterious or possibly damaging by *in silico* tools (SIFT⁽²⁶⁾ and PolyPhen⁽²⁷⁾, respectively) and has a CADD score of 11.91. It is located in exon four out of five and affects a region that is predicted to be intolerant to such substitutions (Figure S5, metadome). It occurs only twice in the gnomAD database (v2.1.1) comprising 141,456 individuals (67,961 XY), once identified in a hemizygous male and once in a female carrier, and is absent from our in house database.

The heterozygous variant c.551G>A in *CYLC2* is predicted to be tolerated (SIFT) or benign (PolyPhen) which is in accordance with a low CADD score of 0.008. It is located in exon five out of eight and affects a region in which variants are likely to be tolerated (Figure S5, metadome). However, it is a rare variant occurring with an allele frequency of 0.0035 in the general population, according to the gnomAD database. Importantly, only three XX individuals are reported to be homozygous for the variant within *CYLC2*.

Segregation analyses revealed maternal inheritance of the X-linked *CYLC1* variant c.1720G>C p. (Glu574Gln), whereas the father carries the heterozygous *CYLC2* variant c.551G>A p. (Gly184Asp) (Fig. 5A). According to ACMG-AMP criteria⁽²⁸⁾ both variants are classified as variants of uncertain significance (Table S2). No other potentially pathogenic variants in genes associated with sperm morphological defects were identified by exploring the exome data of M2270.

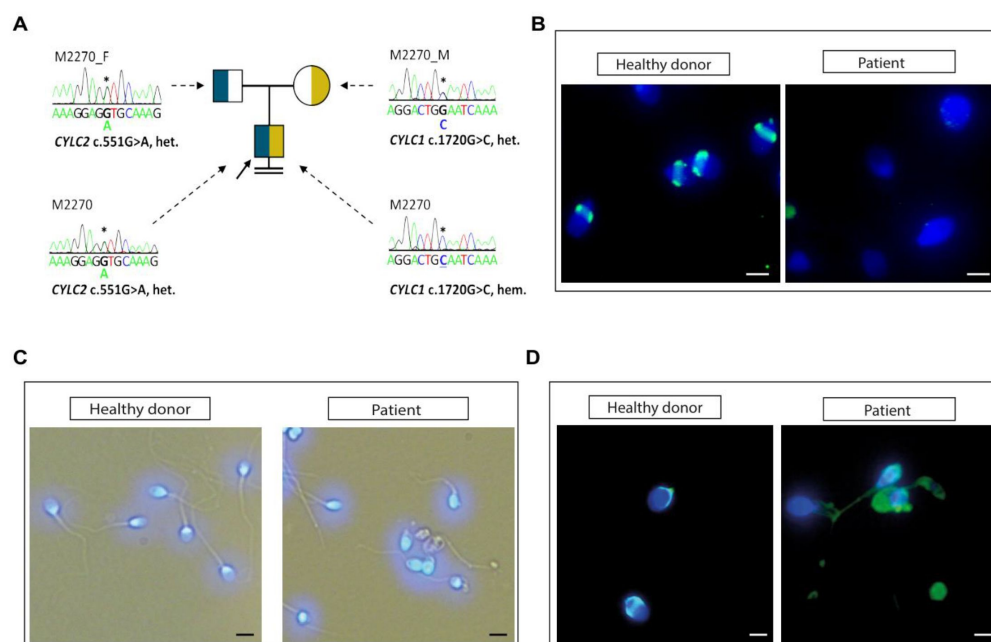


Figure 5:

Cylicins are required for human male fertility

(A) Pedigree of patient M2270. His father (M2270_F) is carrier of the heterozygous *CYLC2* variant c.551G>A and his mother (M2270_M) carries the X-linked *CYLC1* variant c.1720G>C in a heterozygous state. Asterisks (*) indicate the location of the variants in *CYLC1* and *CYLC2* within the electropherograms.

(B) Immunofluorescence staining of *CYLC1* in spermatozoa from healthy donor and patient M2270. In donor's sperm cells *CYLC1* localizes in the calyx, while patient's sperm cells are completely missing the signal.

(C) Bright field images of the spermatozoa from healthy donor and M2270 show visible head and tail anomalies, coiling of the flagellum as well as headless spermatozoa who carry cytoplasmic residues without nuclei. Heads were counter-stained with DAPI.

(D) Immunofluorescence staining of the CCIN in the human sperm cells. While in healthy sperm CCIN locates in the funnel shaped calyx, sperm from M2270 show irregular expression of CCIN and lack of calyx.

Semen analysis performed following WHO guidelines⁽²⁹⁾ is shown in Table 1. The sperm concentration in the semen was slightly reduced, while significant reduction of motile spermatozoa (12.5%) was observed. Interestingly, only 2% of the sperm cells appeared morphologically normal, while 98% of sperm showed head defects (Table 1).

Immunofluorescence staining of *CYLC1* revealed that while in healthy donor's sperm, *CYLC1* localizes in the calyx, in M2270, *CYLC1* labeling was absent (Fig. 5B). Bright field microscopy demonstrated that M2270's sperm flagella coiled similarly to that in Cylicin deficient murine sperm samples. Interestingly, DAPI staining revealed that many flagella carry cytoplasmic bodies without nuclei and could be defined as headless spermatozoa (Fig. 5C). CCIN staining demonstrated that while spermatozoa of a healthy donor showed a typical, funnel-like calyx structure in the posterior region of PT, spermatozoa from M2270 had CCIN localized in irregular manner throughout head and tail, suggesting that Cylicins have a role in maintenance of the calyx structure and composition in both mice and human spermatozoa (Fig. 5D).

Table 1:

Semen analysis of the patient M2770 carrying variants in the *CYLC1* and *CYLC2* genes

	First visit	Second visit	WHO reference rang
Abstinence time [d]	4.0	5.0	
Volume [ml]	4.2	5.8	> 1.4
Concentration [Mill./ml]	10.5	16.3	> 16
Total sperm count [Mill.]	44.1	94.5	> 39
Vitality [%]	53	27	> 54
Motility			
a [%]	7	9	a + b >30
b [%]	5	4	
c [%]	19	8	
d [%]	69	79	
Morphology			
Normal [%]	2	2	> 4
Head defects [%]	99	99	
Midpiece defects [%]	63	59	
Flagella defects [%]	18	47	

Discussion

Spermiogenesis is a highly organized process that is dependent on a unique cytoskeletal organization of the sperm cells. The PT has a role of structural scaffold that surrounds sperm nucleus, and its protein composition is crucial for correct sperm development. In this study, we used CRISPR/Cas9 gene editing to establish *Cylc1*-, *Cylc2*- and *Cylc1/2*-deficient mouse lines to analyze the role of Cylicins. We demonstrated that the loss of Cylicins impairs male fertility in mice by severely disturbing sperm head architecture. The significance of our findings is supported by the identification of *CYLC1/2* variants in an infertile patient who presents similar structural anomalies in sperm cells.

Detailed analysis of sperm development and morphology demonstrated that infertility of Cylicin-deficient male mice is caused by anomalies of different sperm structures. The prominent loss of PT integrity and calyx structure in the mature sperm lead to acrosome loosening and detachment from the nuclear envelope. Furthermore, the shifting of the basal plate caused damage in the head-tail connecting piece resulting in coiling of the flagellum and impaired swimming. Murine Cylicins are first detected in the periacrosomal region of round spermatids and move to the postacrosomal region of PT during spermatid elongation to finally localize to the calyx of mature sperm. This dynamic localization pattern of Cylicins has been described in human, boar and bovine sperm as well ^{(2), (14), (15)}. Interestingly, other PT enriched proteins CCIN ^{(12), (30), (7)} and CPβ3/CPα3 move in the similar manner across PT during spermiogenesis and are present in the calyx of mature sperm of various mammalian species. This co-localization suggests the potential interaction between calyx proteins. Furthermore, CCIN ^{(2), (12), (13)} and CPβ3-CPα3 complex ^{(16)–(18)} are described as actin-binding proteins and porcine *Cylc2* has been shown to have a high affinity for F-actin as well ⁽²⁰⁾. The potential roles of F-actin during spermiogenesis in mammals involve biogenesis of

the acrosome ⁽³¹⁾ and its correct attachment to the outer nuclear membrane of the spermatids ⁽³²⁾ as well as removal of excess cytoplasm ⁽³³⁾.

The loss of Cylicins caused acrosome detachment from NE starting from cap phase of the acrosome biogenesis. Interestingly, loss of CCIN results in similar loosening of the acroplaxome from the outer nuclear membrane ⁽⁷⁾. Calicin is shown to be necessary for the IAM-PT-NE complex by establishing bidirectional connections. Based on our findings, we conclude that both CCIN and Cylicins need to interact in order to provide the ‘molecular glue’ necessary for the acrosome anchoring. Furthermore, Cylicin-deficiency resulted in cytoplasmic retention and bending of the midpiece, similarly to the CPa3 mutant phenotype repro32/repro32 ⁽³⁴⁾. The CAPZ complex has been shown to regulate the actin dynamics by preventing addition or loss of G-actin subunits to the ends of F-actin filaments ⁽³⁵⁾. Since we demonstrated that the loss of Cylicins resulted in mislocalization of both CCIN and CPa3, we speculate that Cylicins have a crucial role in maintenance of the integrity of PT structure and thus are required for proper function of PT proteins.

Other than morphological defects of the mature sperm PT, during spermiogenesis Cylicin-deficiency results in excessive manchette elongation, its delayed disassembly as well as formation of abnormal gaps in the PT at the level of perinuclear ring. The intra-manchette transport (IMT) of proteins from the apical pole of the head to the base of the developing tail occurs during spermiogenesis ⁽³⁶⁾. The malfunctions of the IMT can cause delay in manchette clearance and morphological defects of mature sperm. Mouse models deficient for IMT proteins such as HOOK1 ⁽²⁵⁾ CEP131 ⁽³⁷⁾ and IFT88 ⁽³⁸⁾ show abnormal manchette elongation and delay in its clearance resulting in aberrant nuclear shape of the sperm. Furthermore, the loss of HOOK1 results in head-tail connection and mid-piece anomalies such as flipping of the head, basal plate defects and coiling of the tail similar to *Cylc2*-deficiency ⁽³⁹⁾. Furthermore, we observed wide gaps in the perinuclear ring during manchette elongation, suggesting that Cylicins might have a role in maintenance of the contact between caudal and apical PT region and its integrity.

Our evolutionary analysis of *Cylc1* and *Cylc2* genes across rodents and primates revealed that both coding sequences are under purifying selection. Overall, the results reveal that *Cylc1* is under slightly less conserved constraint than *Cylc2* leading to the hypothesis that loss of function in *Cylc1* might be less severe and could be compensated for by *Cylc2* due to partial redundancy. This hypothesis is supported by our finding that *Cylc1*-deficiency causes subfertility in male mice while the loss of both *Cylc2* alleles results in male infertility. Furthermore, in *Cylc2*^{+/-} male mice fertility was preserved while *Cylc1*^{-/-}*Cylc2*^{+/-} males were unable to sire offspring leading to the conclusion that the loss of one *Cylc2* allele could be compensated by *Cylc1*, however at least two functional Cylicin alleles are required for male fertility in mice.

In general, the evolutionary rate of C-terminal lysine rich region of both Cylicins seems to be highly volatile between conserved and positively selected codon sites, while the lysine residues seem to be strongly conserved. Changes in the C-terminal region, potentially affecting the length of the lysine rich domain, might have an adaptive advantage. Targeted positive selection on codon sites could also be a signature of co-evolution with a rapidly evolving interactor.

Sperm morphological defects and infertility observed in one patient with variants in both Cylicin genes points towards a requirement for human spermiogenesis and fertility. A defect in acrosome formation and the existence of variants in two out of three alleles in both *CYLC* genes is in line with the observations made in mice. The absence of CYLC1 confirmed through immunofluorescent staining indicates a functional impact of the missense variant on the CYLC1 protein. Furthermore, impaired CCIN localization was observed in patient

M2270 as well, suggesting that PT has similar roles in human and rodents despite the differences in sperm head shape⁽⁴⁰⁾. These results suggest that establishing mouse models deficient for Cylicin and other PT proteins might provide insights into mechanisms of human spermiogenesis and cytoskeletal regulation as well. However, with our data we cannot exclude the possibility that there is a discrepancy between mice and men and that the missense variant in *CYLC1* might alone be sufficient to cause the observed phenotype of M2270. The o/e ratios of both genes calculated within the gnomAD database rather indicate a stronger selective pressure on human *CYLC1* (0.08) than on *CYLC2* (0.84). Furthermore, the *CYLC1* variant affects an intolerant region within the protein sequence according to metadome⁽⁴¹⁾, whereas the *CYLC2* variant is located in a region that is more tolerant to variation (Fig. S5). Therefore, we cannot definitively validate the hypothesis of an oligogenic disease in men as well.

The *CYLC2* missense variant is inherited by the father of M2270 (Fig. 5A) and, thus, not sufficient to cause infertility. However, the father reported difficulties to conceive naturally. Based on this one family, we cannot exclude an effect of pathogenic heterozygous *CYLC2* variants.

Overall, the identification and detailed characterization of further patients with variants in *CYLC1* and *CYLC2* is warranted to draw firm conclusions on the effect of variants in these genes on spermiogenesis and infertility.

Supporting information

supplements/521208_file03.pdf Supplementary Materials
supplements/521208_file04.avi
Supplementary Video 1
supplements/521208_file05.avi Supplementary Video 2

Acknowledgements

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Author Contributions

S.S. and H.S. conceived and designed the experiments. S.S. generated gene-edited mouse models. S.S., A.K. and M.M. analyzed *Cylc1* and *Cylc2* deficient mice. L.A. performed evolutionary analyses. A.K.D., S.A.K., C.B. and F.T. analyzed and provided exome sequencing data of infertile men. S.S., A.K. and H.S. were major contributors in writing the manuscript. All authors read and approved the final manuscript.

Methods

Animals

All animal experiments were conducted according to German law of animal protection and in agreement with the approval of the local institutional animal care committees (Landesamt für Natur, Umwelt und Verbraucherschutz, North Rhine-Westphalia, approval IDs: AZ84-02.04.2013.A429, AZ81-02.04.2018.A369). *Cylc1*- and *Cylc2*-deficient mice were generated by CRISPR/Cas9 mediated gene-editing in zygotes of the hybrid strain B6D2F1. Guide sequences are depicted in Table 2. For *Cylc1*, in vitro transcribed sgRNAs (50 ng/μl each) and Cas9 mRNA (100 ng/μl) (Sigma-Aldrich, Taufkirchen, Germany) were microinjected into the cytoplasm of zygotes as described previously⁽⁴²⁾. *Cylc2* deficient mice were generated by electroporation of ribonucleoprotein (RNP) complexes into zygotes using a GenePulser II electroporation device (BioRad, Feldkirchen, Germany). For RNP formation, crRNA and tracrRNA (IDT, Leuven, Belgium) were combined in duplex buffer (IDT) to a final concentration of 50 mM each and annealed (95°C, 5 min; cool down to room temperature with -0,2°C/sec.). For RNP assembly, 4 μM Cas9 protein (IDT) and 2 μM of each annealed cr/tracr RNA were combined in Opti-MEM media (Gibco 11058-021, Thermo Fisher, Carlsbad, USA) and incubated for 10 min at room temperature. RNPs complexes were diluted 1:2 in Opti-MEM, supplemented with 30-40 zygotes in a 0.1 cm gene-pulser cuvette (BioRad) and electroporated (two 30 V square wave pulses, 2 ms pulse length, 100 ms pulse interval). Recovered embryos were cultured over night at 37°C, 5% CO₂ in G-TL medium (Vitrolife, Göteborg, Sweden) and transferred into the oviduct of pseudopregnant CB6F1 foster mice at 2-cell stage. Offspring were genotyped and gene-edited alleles were separated by backcrossing of founder animals with C57Bl/6J mice. Established mouse lines were registered with Mouse Genome Informatics: *Cylc1*^{em1Hsc} (MGI: 7341368), *Cylc2*^{em1Hsc} (MGI:6718280). Mouse lines were maintained as congenic strains on C57Bl/6J background. For all analyses sexually mature males at the age of 8-15 weeks, backcross generation ≥N3 were used.

Table 2:

Name	protospacer sequence (5'-3')
Mm.Cas9.CYLC1.sg1	GGTTTATCCATACGTGAGT
Mm.Cas9.CYLC1.sg2	GGCTTAGGTGATGCTCTAAA
Mm.Cas9.CYLC2.1.AB	AAGGGAGAGTCGAAAAGCGT
Mm.Cas9.CYLC2.1.AF	GGATCCAAGGATAAAGTGTC

Genomic DNA extraction and genotyping

Genomic DNA was extracted from biopsies using the HotShot method ⁽⁴³⁾. PCR reactions were assembled according to the manufacturers protocol of the DreamTaq Green DNA Polymerase (Thermo Fisher, EP0712) using gene specific primers listed in [Table 3](#).

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Table 3:

Cylc1	5'-3'	expected band size
Cylc1_F1	TATACACACAATCCACAATCTTGAAAT	WT: 427 bp
Cylc1_R1	TCACTTCAAAATCCAACCTTGTCT	KO: 264 bp
Cylc1_R2	TGCCTAGTATTTCAAGGTTCCCC	
Cylc2		
Cylc2_F1	ACCACCATTATGGATGCACCG	WT: 376 bp
Cylc2_R1	AGTGTTTCTGTGAGCTCGTTG	KO: 286 bp
Cylc2_R2	GGCTGAATCTTTACCCTTAGGT	

PCR primer sequences

Cylc1	5'-3'	expected band size
Cylc1_F1	TATACACACAATCCACAATCTTGAAAT	WT: 427 bp
Cylc1_R1	TCACTTCAAAATCCAACCTGTCCT	KO: 264 bp
Cylc1_R2	TGCCTAGTATTTTCAGGTTCCCC	
Cylc2		
Cylc2_F1	ACCACCATTATGGATGCACCG	WT: 376 bp
Cylc2_R1	AGTGTTTCTGTGAGCTCGTTG	KO: 286 bp
Cylc2_R2	GGCTGAATCTTTACCCTTAGGT	

Fertility analysis

Males were mated 1:2/1:1 with C57Bl/6J females, which were checked daily for presence of a vaginal plug indicative for successful copulation. Plug positive females were separated and pregnancies as well as litter size were recorded.

Sampling

Mature sperm were obtained from the cauda epididymis in M2 medium (Merck) or PBS preheated at 37°C. Following several incisions of the cauda, sperm were retrieved by flush-out for 15-30 min. The extracted sperm samples were washed in PBS, centrifuged (4000 rpm, 5 min, 4°C) and snap frozen. For squash tubule samples, fresh testes were transferred to PBS and tubules were dissected as described by ⁽⁴⁴⁾. Tubule sections were pressed on a slide, quickly frozen in liquid nitrogen and fixed in 90% ethanol for 5 min.

Semen analysis

Sperm concentrations for at least 4 animals of each genotype were determined using a Neubauer hemocytometer. Viability of sperm was determined by Eosin-Nigrosine staining as described previously ⁽⁴⁵⁾ for at least 3 animals per genotype. Sperm motility was analyzed referring to the WHO guidelines for analysis of human semen. For all analyses at least 100 sperm per individual were analyzed and the percentage of motile/immotile, viable/inviable sperm was calculated.

Quantitative reverse transcription-polymerase chain reaction (qRT-PCR)

RNA was extracted from the testis tissue with TRIzol (Life Technologies, Carlsbad, USA). RNA concentrations and purity ratios were determined by NanoDrop ONE (Thermo Scientific)

measurements. qRT-PCR was performed as described previously ⁽⁴²⁾, on ViiA 7 Real Time PCR System (Applied Biosystems) using Maxima SYBR Green qPCR Mastermix (Thermo Fisher). Beta-actin was used as a housekeeping gene for normalization. Primers used are shown in [table 4](#).

Table 4:

Name	Forward	reverse
Cylc1	GGGGAAAAATAAGCTCATGTGTAG	TTCAGGTTCCCCATTGGTTA
Cylc2	GCATTTCCCAAACCAACAAGG	AACGGATGGTCTCTCGGATATT
beta-actin	TGTTACCAACTGGGACGACA	GGGTGTTGAAGGTCTCAA

High-resolution microscopy of the flagellar beat

Image sequences of mouse sperm were acquired using dark-field at an inverted microscope (IX71; Olympus, Hamburg, Germany), equipped with a dark-field condenser and a high-speed camera (ORCA-Flash4.0 V3, C13220-20 CU, Hamamatsu, Hamamatsu City, Japan). A 10x objective (NA 0.4, UPlanFL; Olympus, Hamburg, Germany) with an additional 1.6x magnifying lens (Olympus, Hamburg, Germany) that was inserted into the light path (final magnification: 16x) was applied. Image sequences were recorded at a rate of 200 frames per second (fps). A custom-made observation chamber was used ⁽²¹⁾. Sperm samples were diluted in THY buffer shortly before insertion of the suspension into the observation chamber.

Sperm nuclear morphology

For the analysis of sperm nuclear morphology, epididymal sperm samples from 3 animals of each genotype were fixed in methanol and acetic acid (3:1). The samples were spread onto a slide and stained with 4',6-diamidino-2-phenylindole (DAPI) containing mounting medium (ROTImount FluorCare DAPI (Carl Roth GmbH, Karlsruhe, Germany)). The sperm cells were imaged at 100x magnification, using a Leica DM5500 B fluorescent microscope. At least 200 pictures were taken from each group and analyzed using Nuclear Morphology software by ⁽⁴⁶⁾ according to the developer's instructions. The minimum detection area was set to 1.000 pixels while the maximum detection area was 7.000 pixels.

Histology

Bouin fixed testis tissues were paraffinized, embedded and sectioned at 3-5 μ m. For histological analysis, the sections were deparaffinized and stained with Periodic acid Schiff (PAS staining). Stained sections were imaged at 20x and 63x magnification under bright field using 5500 B microscope.

qRT primer sequences

Name	Forward	reverse
Cylc1	GGGGAAAAATAAGCTCATGTGTAG	TTCAGGTTCCCCATTGGTTA
Cylc2	GCATTTCCCAAACCAAGG	AACGGATGGTCTCTCGGATATT
beta-actin	TGTTACCAACTGGGACGACA	GGGTGTTGAAGGTCTCAA

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Immunofluorescence

Bouin fixed testis tissue sections were deparaffinized in Xylene and rehydrated in decreasing alcohol to be used for immunohistochemical staining. Squash testis samples fixed in 90% EtOH were used for the staining of α-Tubulin and HOOK1. For sperm immunofluorescence, mature sperm cells isolated from cauda epididymis were fixed with methanol acetic acid (3:1), dropped on glass slides and air dried. After washing in PBS twice, all samples were permeabilized using 0.1% Triton x100 for 10 min at room temperature. The samples were then blocked with 5% BSA for 30 minutes, followed blocking with normal

horse serum (Vectorlabs, Burlingame, USA) for 30 minutes at room temperature. For tissue sections, heat-activated antigen retrieval was performed using buffers at pH6. All primary antibodies were incubated over night at 4°C. Antibodies and dilutions are listed in [table 5](#). The respective secondary antibodies were incubated for 1 h at room temperature using VectaFluor Labeling Kit DyLight 488 and DyLight 594 (Vectorlabs, Burlingame, USA). All slides were mounted with DAPI containing mounting medium (ROTMOUNT FluorCare DAPI, Carl Roth).

Table 5:

Antibody	company	Order-number	Antigen	Dilution IF
a-Tubulin	Merc Milipore (Billerica, USA)	16-232		1:1000
CapZa3	Progen	GP-SH4		1:500
Ccin	Progen	GP-SH3		1:500
Cylc1 (used for mouse)	Davids Biotechnology(Regensburg, Germany)	Custom made polyclonal antibody	AESRKSNDERRKTLKIKFRGK and KDAKKEGKKKGKRESRKKR	1:1000 (sperm cells) 1:500 (testis tissue)
Cylc1 (used for human samples)	Santa Cruz	sc-166400		1:500
Cylc2	Davids Biotechnology(Regensburg, Germany)	Custom made polyclonal antibody	KSVGTHKSLASEKTKKEVK and ESGGEKAGSKKEAKDDKKDA	1:1000 (sperm cells) 1:500 (testis tissue)
Hook1	Proteintech	10871-1-AP		1:500

For the analysis of acrosome biogenesis peanut agglutinin (PNA)-fluorescein isothiocyanite (FITC) Alexa Fluor 488 conjugate (Molecular Probes, Invitrogen, Waltham, USA) was used on the Buin fixed testis tissues. After permeabilization and blocking, the tissues were incubated with PNA-FITC 5 µg/ml for 30 minutes at room temperature. Mature sperm were fixed with paraformaldehyde (4%) for 20 min at room temperature. After PBS washing, the sperm samples were incubated with 5 µg/ml PNA-FITC and 5 nM Mito Red (53271; Sigma Aldrich) for 30 min at room temperature. The slides were then mounted with DAPI mount.

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Transmission electron microscopy

For Transmission Electron Microscopy fresh epididymal sperm and testis tissue were used. After washing with PBS, the samples were fixed in 3% glutaraldehyde at 4°C overnight. The samples were then washed in 0.1 M cacodylate buffer and fixed again in 2% osmium tetroxide at 4% for 2 h. After dehydration in an ascending ethanol row, the samples were contrasted in 70% ethanol 0.5% uranyl acetate for 1.5h at 4°C. Samples were then washed in propylenoxide, three times for 10 minutes at room temperature before embedding in Epon C at 70°C for 48h. Ultrathin sections were examined using a Verios 460L microscope (FEI) with a STEM III-detector.

Evolutionary analysis

Evolutionary rates of mammalian *Cylicin* genes were analyzed according to ⁽⁴⁷⁾. Briefly, *Cylc1* and *Cylc2* nucleotide sequences were obtained from NCBI genbank and Ensembl genome browser. Phylogenetic trees of considered species were built according to the “Tree of Life web project”. The webPRANK software was applied for codon-based alignment of orthologous gene sequences and results were visualized using the ETE toolkit. To determine evolutionary rates of gene sequences across mammals, for different clades and for individual codon sites, the codeml application implemented in the PAML software was used ^{(48), (49)}. The M0 model served as basis for all performed analyses. Different codon frequency settings were tested for the M0 model of each gene and the setting with the highest likelihood was chosen. To test whether alternative models describe the selective constraints within a dataset more precise than the M0 model, likelihood ratio-tests (LRT) were performed. Applied models and LRTs are specified in Table S1.

Study cohort and ethical approval

The MERGE (**M**ale **R**eproductive **G**enomics study) cohort currently comprises over 2,030 men, mainly recruited at the Centre of Reproductive Medicine and Andrology (CeRA) in Münster. The large majority has severe spermatogenic failure, i.e. severe oligozoospermia (<5 Mill./ml sperm concentration), crypto- or azoospermia. So far, only 35 cases were included because of notable sperm morphological defects (≥ 5 Mill./ml sperm concentration, <4 % normal forms). Common causes for infertility such as oncologic diseases, AZF deletions or chromosomal aberrations were ruled out in advance. Patients with aetiologically still unexplained infertility underwent whole exome sequencing.

All participants gave written informed consent according to the protocols approved by the Ethics Committee of the Medical Faculty Münster (Ref. No. MERGE: 2010-578-f-S) in accordance with the Declaration of Helsinki in 1975.

Whole exome sequencing and data analysis

After DNA extraction from patients’ peripheral blood lymphocytes WES was performed as previously described ⁽⁵⁰⁾. WES data obtained from 2,066 infertile men was filtered for rare (≤ 0.01 minor allele frequency, gnomAD) variants located within the coding sequence or the adjacent 15 bp of flanking introns in *CYLC1* and *CYLC2*. Patients carrying only one variant in either of the gene were excluded. In case of either bi-allelic *CYLC2* variants or a combination of *CYLC1* and *CYLC2* variants, the whole exome dataset was screened to rule out other potential genetic causes. Variants detected in this study were classified according to the guidelines by the American College of Medical Genetics and Genomics-Association for Molecular Pathology (ACMG-AMP) ⁽²⁸⁾ adapted to recent recommendations as outlined in ⁽⁵¹⁾.

To rule out an alternative genetic cause for the patient’s condition, his exome data was screened for rare ($MAF \leq 0.01$, gnomAD and in-house database), homozygous, X-linked or potentially compound heterozygous high-impact variants (stop gain, start lost, stop lost, frameshift, splice site and splice region as well as missense variants with CADD > 20) and rare, heterozygous LoF variants (stop gain, start lost, stop lost, frameshift, splice site) without filtering for a specific set of genes. Respective genes were screened for testis-expression and reports of infertility.

Sanger sequencing

Sanger sequencing with variant specific primers was conducted for validation and segregation purposes. The primers to confirm the *CYLC1* (NM_021118.3) variant c.1720G>C p.(Glu574Gln) are 5'-ACTGATGCTGACTCTGAACCG-3' (forward) and 5'-CCTTCGAGTCACAAAGTTGTCA-3' (reverse). To confirm the *CYLC2* (NM_001340.5) variant c.551G>A p.(Gly184Asp), the primers 5'-CTGTCGAGGTGGATTCTAAAGC-3' (forward) and 5'-TGCATCCTTCTTGCATCCT-3' (reverse) were used.

Analysis of the human sperm samples

Human ejaculate samples from healthy donor and patient M2270, were analyzed according to WHO guidelines prior to washing in buffer and centrifuged (1000 rpm, 20 min). The cells were fixed in methanol and acetic acid (3:1) and used for immunofluorescence staining. Samples were dropped on slides and permeabilized with 0.1% Triton x100. After blocking with 5% BSA for 30 min, slides were incubated with primary CCIN or CYLC1 antibodies (concentrations shown in Table 3) for 3 h at room temperature. Secondary antibodies were incubated for 1 h, followed by mounting with DAPI containing medium.

Statistics

For all analyses mean values +/- SD were calculated. Statistical significance was determined by One-way ANNOVA analyses using Bonferroni correction.

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Reviewer #1 (Public Review):

Mice and humans have two Cylicin genes (X-linked Cylicin 1 and the autosomal Cylicin 2) that encode cytoskeletal proteins. Cylicins are localized in the acrosomal region of round spermatids, yet they resemble a calyx component within the perinuclear theca of mature sperm nuclei. The function of Cylicins during this developmental stage of spermiogenesis (tail formation and head elongation/shaping) was not known. In this study, using CRISPR/Cas genome editing, the authors generated *Cylc1*- and *Cylc2*-knockout mouse lines to study the loss-of-function of each Cylicin or all together.

The major strengths of the study are the rigorous and comparative phenotypic analyses of all the combinatorial genotypes from the cross between the two mouse lines (*Cylc1*^{-/-}, *Cylc2*^{-/-}, *Cylc1*^{-/-} *Cylc2*^{+/-} and *Cylc1*^{-/-} *Cylc2*^{-/-}) at the levels of male fertility, cellular, and subcellular levels to support the conclusion of the study. While spermatogenesis appeared undisturbed, with germ cells of all types detected in the testis, low sperm counts in epididymis were observed. Mice were subfertile or infertile in a dose-dependent manner where fewer functional alleles had more severe phenotypes; the loss of *Cylc2* was less tolerated than the loss of *Cylc1*. Thus, loss of *Cylc1*, and to an even greater extent, loss of *Cylc2*, leads to sperm structure anomalies and decrease sperm motility. Particularly, the sperm head and sperm head-neck region are affected, with calyx not forming in the absence

of Cylicins, the acrosomal region being attached more loosely, and the sperm head itself appearing structurally rounder and shorter. Furthermore, manchette, which disassembles during spermiogenesis, persists in mature sperm of mice missing *Cylc2*. It is interesting that the study identifies a human male that has mutations in both *CYLC1* and *CYLC2* genes, and suffers from infertility, with similar motility and sperm structure defects compared to the mouse models. *CYLC1* in the sperm from the infertile patient is absent, providing evidence that in both rodents and primates, Cylicins are essential for male fertility.

The major weakness of the study is the less robust or absent of statistical analyses determining the statistical significance of some of the morphological phenotypes observed (e.g., the roundness/shortening of sperm head). Evolutionary analysis of two genes-while interesting- is less congruent with the other parts of the study and disrupts the overall flow of the functional studies. The authors show that the reason for the loss of *Cylc2* being more severe is due to the higher conservation of *Cylc2* compared to *Cylc1* in rodents and primates, however, the conservation of these genes in other species is not discussed.

Overall, the work highlights the relevance and importance of Cylicins in male infertility and advances our understanding of perinuclear theca formation during spermiogenesis.

Reviewer #2 (Public Review):

The work presented in this manuscript focuses on the role of Cylicins in spermiogenesis and the consequences of their absence on infertility. The manuscript is presented in two parts: the first part studies the absence of Cylicins from KO mouse models and shows in mice that both isoforms of Cylicins are necessary for normal spermiogenesis. The evaluation of double heterozygotes is particularly useful for the second part which looks at the presence of mutations in these genes in a cohort of infertile men. A patient with two hemizygous/heterozygous mutations in the *CYLC1* and 2 genes, respectively, was identified for the first time and the results obtained with the KO models support the hypothesis of the pathogenicity of the mutations.

In general, the experiments are perfectly performed and the results are clear. Numerous techniques in the state of the art in male reproduction are used to obtain high-quality phenotyping of the mouse models.

The discovery of two concomitant mutations in an infertile patient is very interesting and the work carried out on mice allows supporting that an absence of *CYLC1* and a heterozygous mutation of *CYLC2* could lead to a phenotype of complete infertility. However, as the mutation on *CYLC2* is not identified as pathogenic, the pathogenicity of this mutation remains in question (the authors note this point in the discussion). It would be interesting to see if the mutated amino acid is conserved between different species. In mice, the authors have shown the importance of these proteins on the morphology of the acrosome. What about in humans?

Reviewer #3 (Public Review):

The authors tried to study the role of the cylicin gene in sperm formation and male fertility. They used the Crispr/cas 9 to knockout two mouse cylicin genes, cylicin 1 and cylicin 2. They used comprehensive methods to phenotype the mouse models and discovered that the two genes, particularly cylicin 2 are essential for sperm calyx formation. They further compared the evolution of the two genes. Finally, they identified mutations of the genes in a patient. The major strengths are the high quality of data presented, and the conclusion is supported by their findings from the animal models and patients. The major weakness is that the study

is descriptive: no molecular mechanism studies were conducted or proposed, limiting its impact on the field.